

**PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE
(IOHE)**

On

**OPTIMIZING EMOTION RECOGNITION USING EFFICIENT
NUREAL NETWORK**

**Submitted in partial fulfillment of the requirements for the award of
degree of**

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted By:

Aditya Bisht

2210990061

Supervised By:

Dr. Gifty Gupta

Mentor G-18



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**CHITKARA UNIVERSITY, INSTITUTE OF ENGINEERING AND
TECHNOLOGY**

CHITKARA UNIVERSITY, PUNJAB, INDIA

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **“Optimizing Emotion Recognition Using Efficient Neural Network”** in partial fulfillment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Aditya Bisht

Date: 5/13/2026

2210990061

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Gifty Gupta

Mentor G-18

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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ABSTRACT

Facial emotion recognition (FER) has emerged as an important research area in computer vision and affective computing due to its applications in human–computer interaction, healthcare systems, intelligent surveillance, driver monitoring systems, and affect-aware technologies. The ability of machines to automatically identify human emotions through facial expressions has significantly improved with the advancement of deep learning techniques, particularly convolutional neural networks (CNNs). However, many high-performing deep learning models require substantial computational resources and memory, limiting their deployment in real-time and resource-constrained environments.

This project focuses on optimizing emotion recognition using an efficient convolutional neural network architecture designed for real-time facial emotion classification. The proposed model is trained and evaluated using the FER-2013 dataset under consistent experimental conditions. Various preprocessing and augmentation techniques are applied to improve model generalization and reduce overfitting. The performance of the model is evaluated using classification metrics such as accuracy, precision, recall, and F1-score. In addition, training behavior is analyzed using accuracy and loss curves to better understand model convergence and learning performance.

The experimental results demonstrate that the proposed approach achieves strong emotion recognition performance while maintaining practical computational efficiency. The findings emphasize the importance of balancing predictive accuracy with deployability when designing real-world FER systems. Overall, the study provides valuable insights into the development of efficient and scalable deep learning models for real-time emotion recognition applications.

1. Introduction

Facial emotion recognition (FER) is an important research area in computer vision and affective computing. It enables machines to identify human emotions through facial expressions, improving human–computer interaction systems.

Emotion recognition systems are widely used in healthcare, intelligent surveillance, driver monitoring systems, virtual assistants, and educational applications. Recent developments in deep learning and convolutional neural networks (CNNs) have significantly improved the performance of FER systems.

Despite these advancements, many deep learning models require high computational power and memory, making them difficult to deploy in real-time environments or on resource-constrained devices.

This project focuses on developing an efficient neural network model capable of achieving high accuracy while maintaining computational efficiency for practical deployment.

1.1 Objectives

- To develop an efficient CNN-based emotion recognition model.
- To perform facial emotion classification using the FER-2013 dataset.
- To improve model generalization using preprocessing and augmentation techniques.
- To analyze model performance using classification metrics.
- To design a model suitable for real-time deployment.

2. Methodology

The proposed methodology consists of dataset preprocessing, augmentation, model development, training, and evaluation.

2.1 Dataset

The FER-2013 dataset is used for training and evaluation. The dataset contains grayscale facial images of size 48×48 pixels categorized into seven emotion classes:

- Angry
- Disgust
- Fear
- Happy
- Sad
- Surprise
- Neutral

The dataset includes variations in lighting conditions, facial orientations, and expression intensity, making the classification task challenging.

2.2 Data Preprocessing and Augmentation

To improve model performance and reduce overfitting, preprocessing and augmentation techniques are applied.

- Preprocessing Techniques
 - Image normalization
 - Rescaling pixel values
 - Label encoding
 - Dataset splitting
- Augmentation Techniques
 - Horizontal flipping
 - Rotation
 - Zoom transformations
 - Image shifting

These techniques improve data diversity and enhance model generalization.

2.3 Model Architecture

The proposed model is based on a convolutional neural network architecture designed for efficient feature extraction.

- The architecture includes:
- Convolutional layers
- Batch normalization
- Max pooling layers
- Dropout layers
- Fully connected dense layers
- Softmax output layer

The model extracts hierarchical facial features while maintaining computational efficiency suitable for real-time applications.

2.4 Model Training Procedure

The model is trained using a supervised learning approach. The categorical cross-entropy loss function is used for multi-class classification.

- Training Configuration
- Optimizer: Adam
- Batch Size: 64
- Learning Rate: 0.001
- Epochs: 40–50
- Loss Function: Categorical Cross-Entropy

Dropout and validation monitoring are used to reduce overfitting and improve generalization.

2.5 Evaluation Metrics

The performance of the model is evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics provide a comprehensive evaluation of classification performance.

3. Tools and Technology

The project is implemented using the following tools and technologies:

Tools/Technology	Purpose
Python	Programming Language
TensorFlow	Deep Learning Framework
Keras	Neural Network API
Jupyter Notebook	Development Enviornment
NumPy	Numerical Computation
OpenCV	Image Processing
Matplotlib	Visualization
Scikit-learn	Performance Evaluation

4. Implementation

The implementation phase includes dataset loading, preprocessing, augmentation, model building, training, and testing.

The FER-2013 dataset is first loaded and preprocessed. Data augmentation techniques are then applied to improve dataset diversity.

A CNN-based architecture is developed using TensorFlow and Keras. The model is trained using the Adam optimizer and categorical cross-entropy loss function.

Training and validation accuracy/loss are monitored during the training process. After training, the model is evaluated using classification metrics and confusion matrix analysis.

5. Major Findings/Results

The proposed model achieved strong emotion classification performance on the FER-2013 dataset.

5.1 Training Performance

The training and validation accuracy increased consistently during training, while loss values decreased steadily, indicating effective learning behavior.

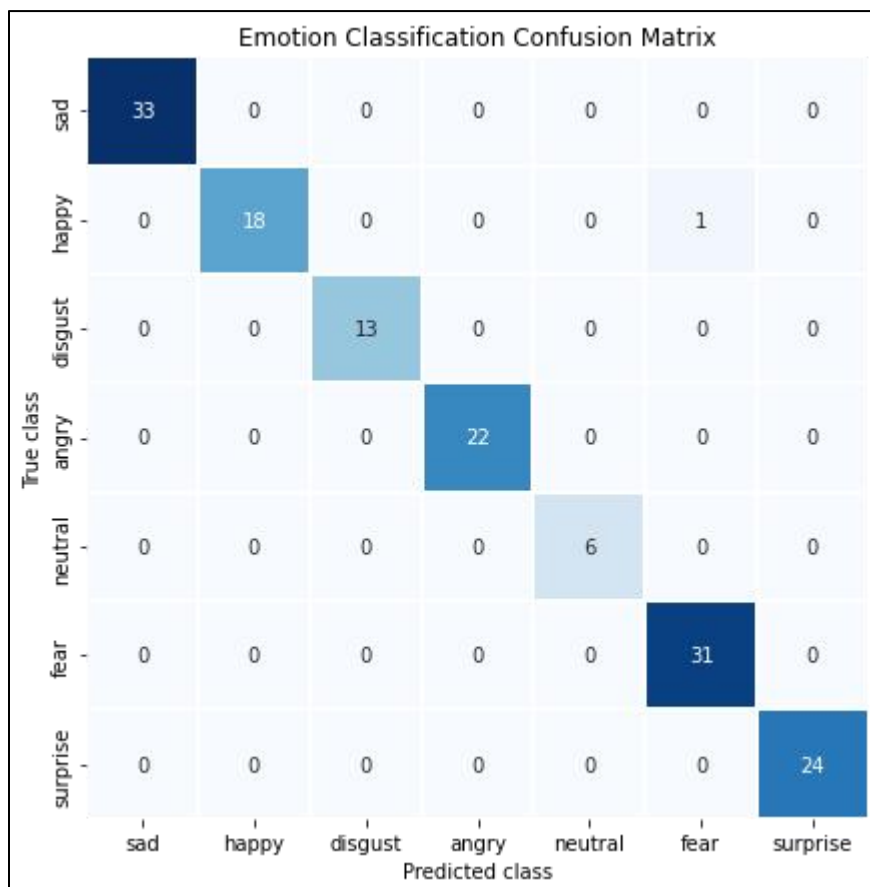
5.2 Classification Performance

The model demonstrated strong performance across multiple emotion categories using metrics such as accuracy, precision, recall, and F1-score.

	precision	recall	f1-score	support
sad	1.00	1.00	1.00	33
happy	1.00	0.95	0.97	19
disgust	1.00	1.00	1.00	13
angry	1.00	1.00	1.00	22
neutral	1.00	1.00	1.00	6
fear	0.97	1.00	0.98	31
surprise	1.00	1.00	1.00	24
accuracy			0.99	148
macro avg	1.00	0.99	0.99	148
weighted avg	0.99	0.99	0.99	148

5.3 Confusion Matrix Analysis

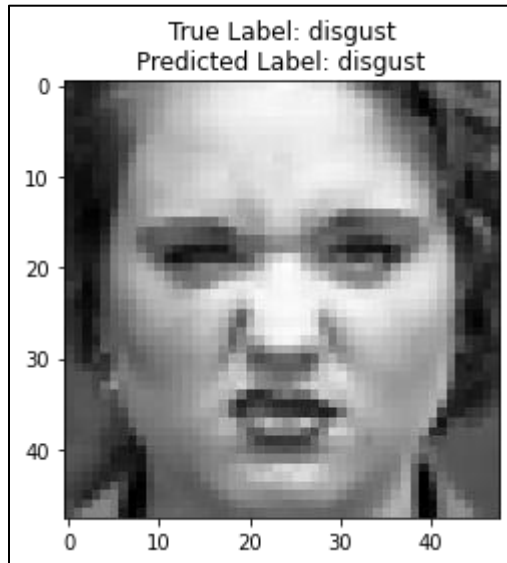
The confusion matrix showed strong diagonal values, indicating correct predictions for most emotion classes. Minor misclassifications were observed between visually similar emotions.



5.4 Efficiency-Oriented Performance Analysis

The proposed model balances predictive accuracy and computational efficiency, making it suitable for real-time emotion recognition systems and deployment in resource-constrained environments.

ResNet Model - Accuracy: 98.65% | Loss: 0.1088



6. CONCLUSION AND FUTURE SCOPE

This project presented an efficient deep learning approach for facial emotion recognition using convolutional neural networks and the FER-2013 dataset.

The proposed model successfully achieved high classification performance while maintaining practical computational efficiency suitable for real-time applications.

The study demonstrated the importance of balancing model accuracy and efficiency when designing real-world emotion recognition systems.

6.1 Future Scope

Future improvements may include:

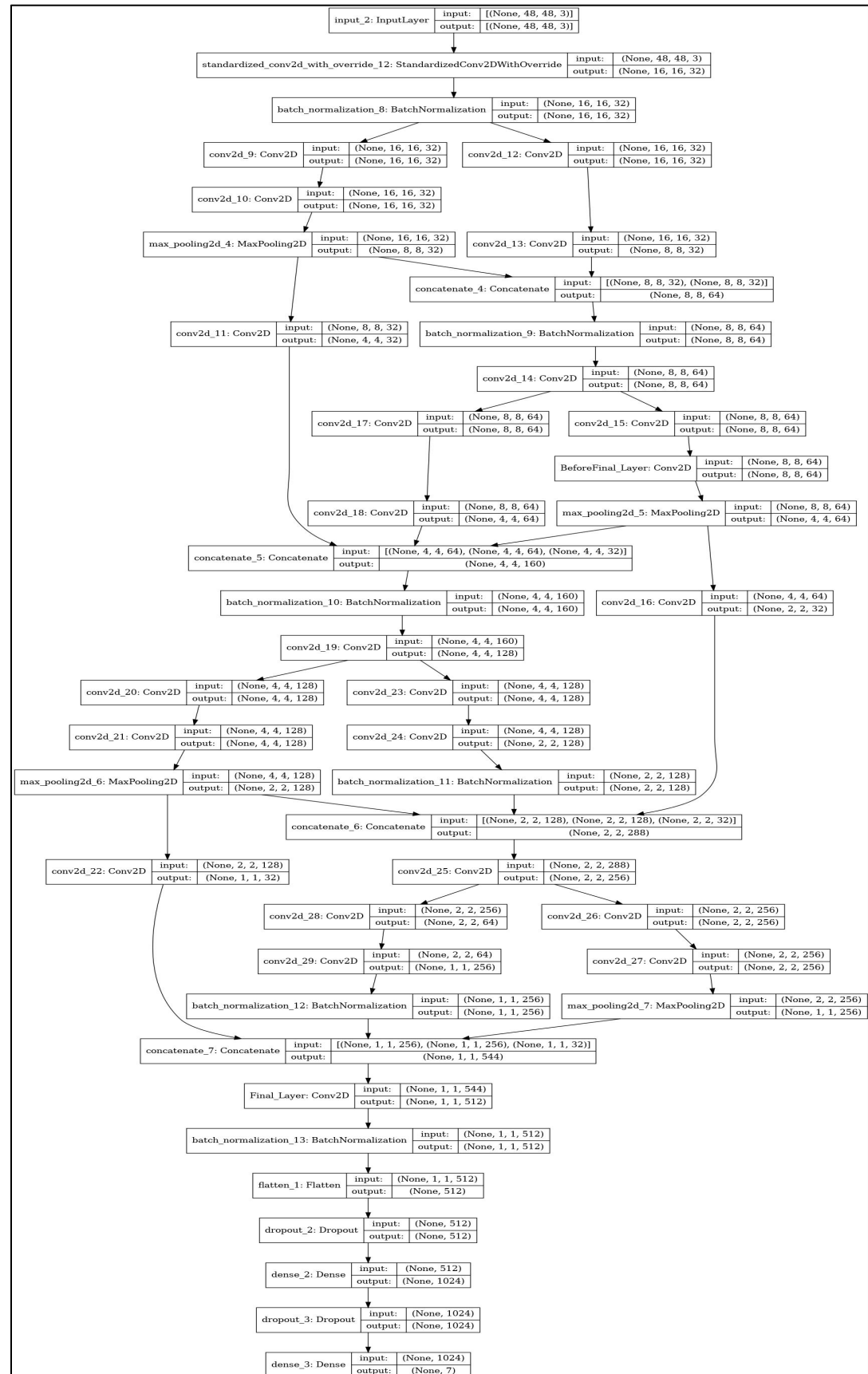
- Integration with live webcam-based emotion recognition systems.
- Deployment on mobile and embedded devices.
- Use of transfer learning and transformer-based architectures.
- Real-time emotion analytics applications.
- Multimodal emotion recognition using audio and facial cues.

References

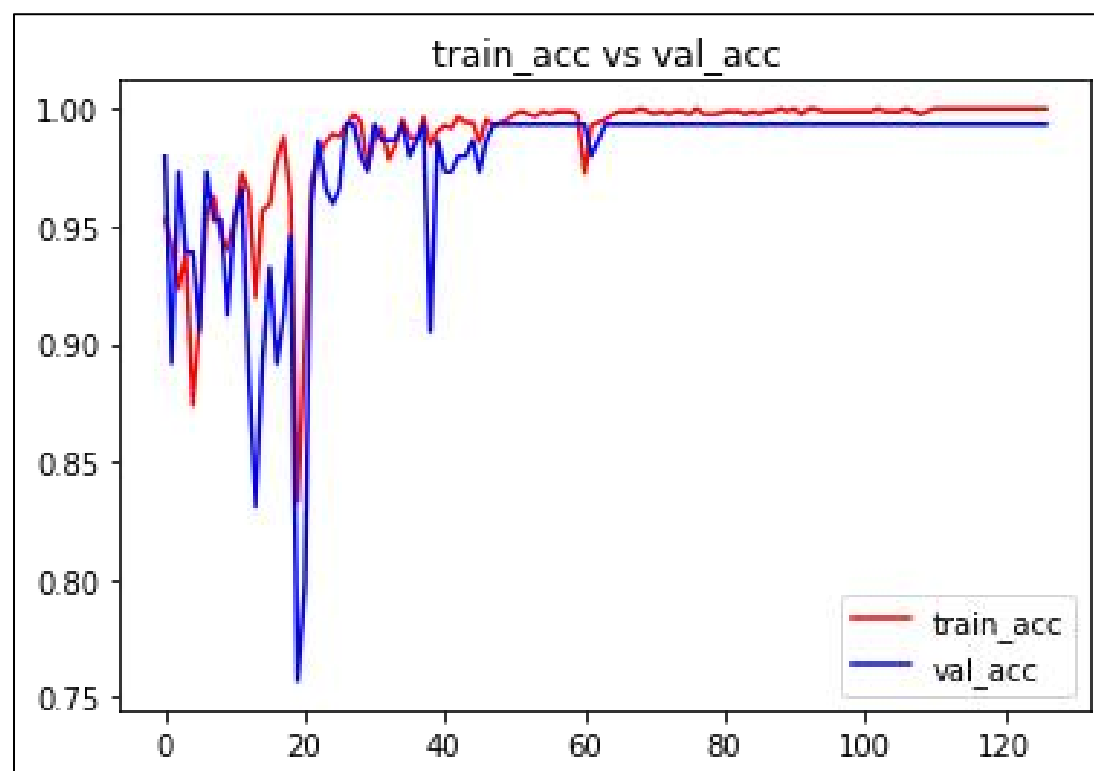
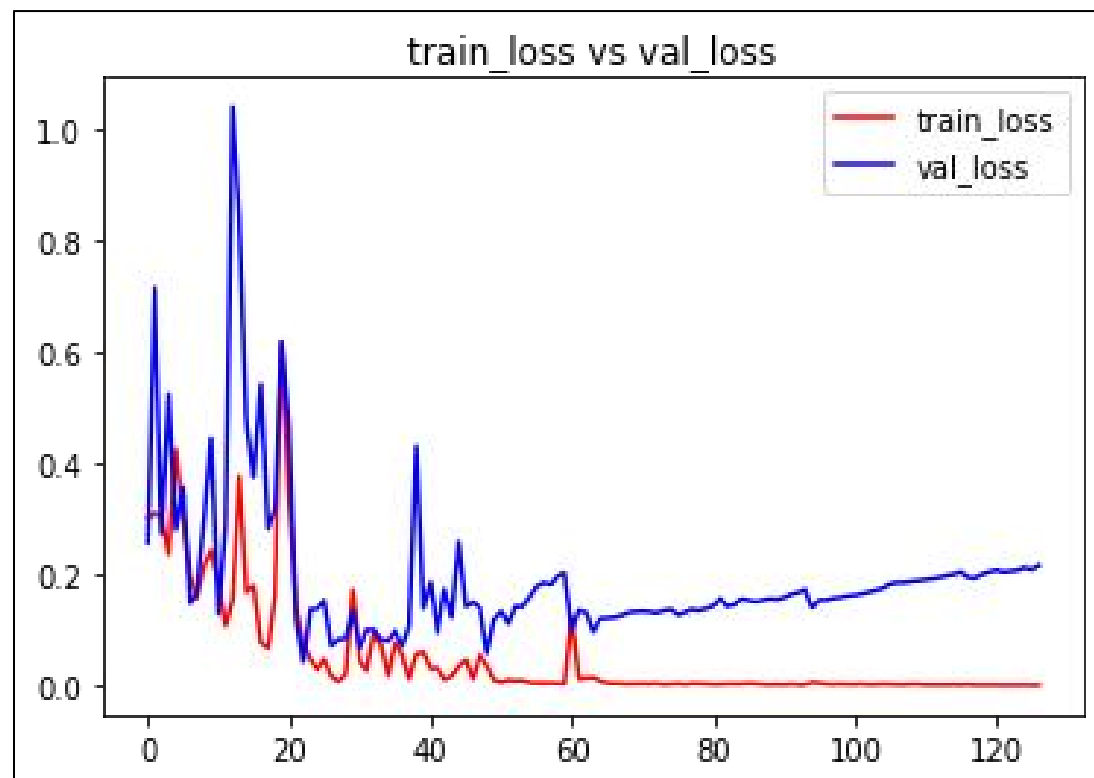
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APPENDICES

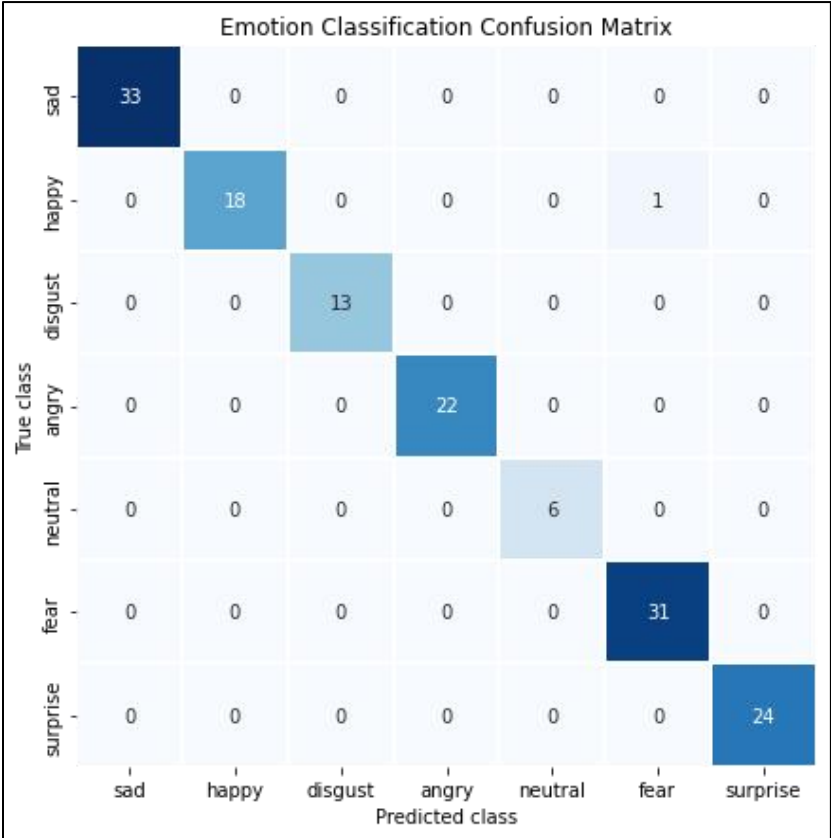
● Model Architecture Diagram



- Training Accuracy Graph



● **Confusion Matrix**



● **Dataset Images**

